KHYATI KIYAWAT

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Research Focus

Interests: Memory-Centric Architecture, LLM Inference Acceleration, Energy- and Cost-Efficient Systems

My research addresses the memory bottleneck in LLM inference by co-designing processing-in-memory (PIM) and processing-near-memory (PNM) systems, including chiplet-based accelerators and computational CXL devices, with a focus on developing performance modeling frameworks and tools.

Education

University of Virginia, VA, USA 2022–2028 (expected)
Ph.D., Department of Computer Science (Advised by Prof. Kevin Skadron) GPA: 3.88/4.0

Indian Institute of Technology Roorkee, India 2016–2020
B.Tech., Department of Electronics and Communication GPA: 9.23/10.0

Industry Experience

Memory Solution Research Intern | Samsung Semiconductors, USA May 2026–Aug 2026
Mentors: Gongjin Sun, Caroline Kahn

- Characterized the performance of Samsung's CXL-based memory devices as a KV cache offload tier for LLM inference using LMCache and compared them against System DDR and SSD baselines. (US patent to be filed)

Processor Architecture Research Intern | Intel, India May 2024–Jan 2025
Mentors: Shankar Balachandran, Anant V. Nori, Sreenivas Subramoney

- Built a tensor-tracking framework for GenAI workloads that enables rapid exploration of lossless KV cache compression in Xeon LLCs.

Front-end Design Verification Engineer | Texas Instruments, India Summer 2019, Jul 2020–Jul 2022

- Verified SoC integration of registers, communication peripherals, and co-processors for C2000 real-time MCUs.
- Developed a Perl-based automation framework to auto-generate RTL for memory-mapped registers and the verification test cases, significantly cutting manual effort and register-RTL errors by 80%.

Publications

[9] **Trace-Driven Analysis for KV Cache Tier Design**
Khyati Kiyawat, Kevin Skadron. *IEEE CAL, 2026 (under review).* [link]

[8] **Reimagining LLM Inference Infrastructure with Memory-Centric KV Cache Servers**
Khyati Kiyawat, Kevin Skadron. *HotInfra, 2026 (co-located with ISCA 2026).* [link]

[7] **HARMONI: Hierarchical ARchitecture MOdeling for LLMs with Near/In Memory Computing**
Khyati Kiyawat, Yasas Seneviratne, Zhenxing Fan, Morteza Baradaran, Kevin Skadron.
ISPASS, 2026. [link]

[6] **Sangam: A Chiplet-Based DRAM-PIM Accelerator with CXL Integration for LLM Inferencing**
Khyati Kiyawat, Zhenxing Fan, Yasas Seneviratne, Morteza Baradaran, Akhil Shekar, Zihan Xia, Mingu Kang, Kevin Skadron. *arXiv preprint, Nov 2025 (under review).* [link]

[5] **TriPIM: Exact Triangle Counting on UPMEM PIM for Graph Analytics**
Morteza Baradaran, **Khyati Kiyawat**, Akhil Shekar, Abdullah Mughrabi, Kevin Skadron.
LBR MemSys, 2025. [link]

[4] **Membrane: Accelerating Database Analytics with DRAM-Based PIM Filtering and Schema Denormalization**
Akhil Shekar, Kevin Gaffney, Martin Prammer, **Khyati Kiyawat**, Lingxi Wu, Helena Caminal, Zhenxing Fan, Yimin Gao, Ashish Venkat, José F. Martínez, Jignesh Patel, Kevin Skadron. *TACO, 2025.* [link]

[3] **Architectural Modeling and Benchmarking for Digital DRAM PIM**
F. Siddique, D. Guo, Z. Fan, M. Gholamrezaei, M. Baradaran, A. Ahmed, H. Abbot, K. Durrer, K. Nandagopal, E. Ermovick, **K. Kiyawat**, B. Gul, A. Mughrabi, A. Venkat, K. Skadron. *IISWC, 2024.* [link]

[2] **An Efficient Scaling-Free Folded Hyperbolic CORDIC Design Using a Novel Low-Complexity Power-of-2 Taylor Series Approximation**
Anu Verma, **Khyati Kiyawat**, Bishnu P. Das, Pramod K. Meher. *TVLSI, 2023.* [link]

[1] **Real-Time Minimum Energy Point Tracking Using a Predetermined Optimal Voltage Setting Strategy**
Khyati Kiyawat, Yutaka Masuda, Jun Shiomi, Tohru Ishihara. *ISVLSI, 2020.* [link]

Teaching and Mentoring

Graduate Teaching Assistant | University of Virginia, USA
- Computer Systems and Organization I (CS 2130) *Fall 2023*
- Computer Systems and Organization II (CS 3130) *Spring 2024*
- CPU/GPU Memory Systems (CS 6501 Special Topics) *Spring 2025, Spring 2026*

Graduate Mentorship Programs | University of Virginia, USA 2024–present
- Mentoring an undergraduate student on KV cache optimization research for Agentic AI (with Dr. Skadron).
- Mentored a junior PhD student on CS doctoral program navigation (CS Graduate Student Group).
- Mentored a CS undergraduate on graduate school applications (Engineering Grad School Mentor Program).

Rise Together | Charlottesville, Virginia Oct 2023–Apr 2024
- Assisted high-school students in the US with undergraduate college applications and the transition process.

Student Mentorship Programme | IIT Roorkee, India Aug 2018–May 2020
- Mentored 12 first-year students in their transition and adjustment to the institute's culture.

Honors and Awards

- **Most Memorable Internship Project** | Samsung Semiconductors Intern Showcase *Aug 2026*
- **Best Demo Award** | JUMP 2.0 PRISM Annual Review *Nov 2025*
- **Selected Participant** | CRA-WP Grad Cohort Workshop for Women *2023*
- **Engineering Distinguished Fellowship** | University of Virginia, awarded to top Ph.D. applicants *2022*
- **Semi-finalist** | Swadeshi Microprocessor Challenge, RISC-V autonomous distress system *2020–21*
- **Honda Y-E-S Award** | 1 of 14 recipients across 6 IITs; Y-E-S Plus Scholarship *2018–19*
- **Quarter-finalist** | TI India Innovation Challenge Design Contest (IICDC) *2018*
- **1988 Batch Award** | IIT Roorkee, top undergraduate by academic merit *2017*

Service

Artifact Evaluation Committee | ISCA 2024, 2025
- Reviewed research artifacts accompanying accepted papers.

Steering Committee Member | Computer Architecture Student Association (CASA) 2025
- Leading the Computer Architecture Long-term Mentorship (CALM) program.

Systems Interest Group (SIG) | University of Virginia Nov 2022–May 2024
- Organized cross-departmental systems research forum, facilitating faculty-student discussions.

Community Outreach | IIT Roorkee, India
- **Unnat Bharat Abhiyan**: Surveyed farming challenges in partner villages under a national program that pairs IITs with rural communities, and designed a sensing system to protect farms from wild animals *(2016–18)*.
- **SOPAN**: Ran weekly outreach activities for primary and middle school students in rural Uttarakhand *(2018–20)*.

Presentations

- Reimagining LLM inference with Memory-centric Servers at *ORNL Core Universities AI Workshop 2026*.
- Open-Source Processing-In-Memory (PIM) Architecture Design through FPGA Emulation: A Case Study Modeling Sieve at *Open-Source Computer Architecture Research Workshop* [OSCAR @ ISCA 2023].
- PIMperf: Analytical Modeling for SIMD Instructions on PIM at *CAPWIC 2024*.
- Chiplet-based DRAM PIM Architecture for LLM Inference at *TECHCON 2024*.